

Multiple Choice (10 marks)

1. A ball is thrown downward onto a concrete floor and bounces upward. What supplies the upward force that makes the ball bounce?
 - (a) the floor
 - (b) the pull of gravity
 - (c) the air friction on the ball
 - (d) the person that throws the ball
2. Which of the following is an acquired human characteristic?
 - (a) Eye color
 - (b) Hair color
 - (c) Height
 - (d) Verbal accent
3. The amount of gravitational attraction between the Moon and an astronaut standing on its surface depends on the astronaut's distance from the center of the Moon and the astronaut's
 - (a) height.
 - (b) mass.
 - (c) shape.
 - (d) weight.
4. When it rains, some animals will _____.
 - (a) hibernate for the season
 - (b) migrate to warmer climates
 - (c) change their body covering
 - (d) move to seek shelter
5. Which method is the safest way to watch an eclipse of the Sun?
 - (a) Turn away after two or three minutes.
 - (b) Look at the Sun through a long telescope.
 - (c) Cast an image through a pinhole onto a screen.
 - (d) Blink often until your eyes get used to the light.

True / False (10 marks)

Answer True or False

6. True or False: When a bird drops a seed in soil, it increases the seed's chances of reproducing into a new plant.
7. True or False: A simple machine changes the weight of heavy objects, making them lighter to lift.
8. True or False: A rabbit's fur turning white in winter is an adaptation to habitat destruction rather than seasonal change.
9. True or False: A boulder sitting on a cliff represents kinetic energy changing to potential energy.
10. True or False: Coal is considered a renewable resource because it can be replenished over time through natural processes.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the various factors that influence a farmer's choice of crops to grow, and analyze how these factors can impact agricultural practices and sustainability.
12. Discuss why temperature changes along coastlines are generally less extreme than those experienced further inland, providing scientific explanations for this phenomenon.

End of Paper